OCEANSENSE: AN IOT-BASED HYBRID SYSTEM FOR REAL-TIME DETECTION AND TRACKING OF MARINE DEBRIS

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Project Individual Summary Report

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(Ms. Narmada Gamage)

Abstract

Marine debris monitoring presents significant challenges due to the dynamic and unpredictable nature of ocean environments. This individual research component, part of the broader OceanSense project, focuses on the design and implementation of a decentralized, environment-aware swarm coordination system for Autonomous Surface Vehicles (ASVs). The ultimate goal of the OceanSense project is to deploy a fleet of *physical*, low-cost IoT sensor nodes capable of real-time marine debris detection and tracking. Within that goal, this component specifically addresses the swarm coordination layer: the algorithms that govern how physical ASVs collaborate and adapt their formation in response to dynamic ocean conditions.

The key novelty lies in using real-time local environmental state variables — including wave height, current direction, and debris density — to drive swarm formation and movement decisions in a fully decentralized manner. Unlike conventional swarm systems that operate on fixed formations or ignore environmental context, this work embeds adaptive coordination logic directly into each node's control algorithm. Nodes dynamically adjust their inter-vehicle spacing, orientation, and search patterns in response to measured conditions, with all coordination performed via minimal LoRa message exchanges.

The coordination logic and algorithms are validated through simulation using Webots and ROS, with realistic models of ocean conditions including wave fields, currents, and sparse LoRa communication. This simulation-based validation phase serves as the precursor to physical hardware deployment. Performance is evaluated on metrics such as area coverage efficiency, formation stability, LoRa message delivery rate, node-failure recovery time, and energy consumption. This research contributes a scalable and resilient swarm coordination approach suitable for persistent, infrastructure-free marine debris monitoring on physical ASV platforms.

Keywords: Swarm Robotics, ASV, LoRa, Decentralized Coordination, Marine Debris, IoT, Environment-Aware Control, Physical Node Deployment

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Finally, I wish to thank my family and friends for their unwavering encouragement and support throughout my academic journey.

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List of Abbreviations

ASV	Autonomous Surface Vehicle
AUV	Autonomous Underwater Vehicle
DC	Decentralized Coordination
FSM	Finite-State Machine
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IoT	Internet of Things
LPWAN	Low-Power Wide-Area Network
LoRa	Long Range (radio communication technology)
LoRaWAN	LoRa Wide Area Network
MAC	Medium Access Control
ML	Machine Learning
MPC	Model Predictive Control
PID	Proportional-Integral-Derivative (controller)
RL	Reinforcement Learning
ROS	Robot Operating System
RSSI	Received Signal Strength Indicator
SI	Swarm Intelligence
SLAM	Simultaneous Localization and Mapping
USV	Unmanned Surface Vehicle

Introduction

1.1 Background & Literature Survey

Marine debris, particularly plastic pollution, has reached crisis proportions globally, with an estimated eight million metric tons of plastic entering the oceans annually [1]. Traditional monitoring approaches — satellites, manned vessels, and stationary sensor buoys — are either prohibitively expensive, limited in spatial coverage, or require extensive human oversight. The growing urgency of this environmental problem demands affordable, scalable, and autonomous monitoring solutions capable of persistent operation across large and dynamic oceanic areas.

Unmanned Surface Vehicles (USVs) and Autonomous Surface Vehicles (ASVs) have emerged as a highly promising platform for environmental monitoring. Their ability to cover diverse aquatic areas with greater spatial resolution and operational flexibility compared to fixed sensing networks makes them well-suited for debris detection tasks [2]. When organised into coordinated swarms, these vehicles can collectively survey large regions, adapt to environmental changes, and maintain continuous coverage even in the event of individual node failures. Swarm robotics approaches, inspired by collective behaviour in nature, have demonstrated high scalability and fault tolerance in such open-world settings [3].

Communication in marine environments is a critical constraint for any multi-robot system. LoRa (Long Range) radio technology has emerged as a practical and energy-efficient solution for long-range, low-power coordination between IoT nodes. LoRa radios provide communication ranges of several kilometres with minimal power consumption, though at the cost of severely limited data rates (typically in the hundreds of bits per second) and mandatory duty-cycle restrictions [4]. Recent work by Souli et al. [5] demonstrated the viability of ROS-LoRa integrated UAV swarms for disaster response, confirming that LoRa-based coordination is feasible even in challenging operational contexts. Similarly, the Golden Seal project [6] demonstrated the use of LoRa-equipped IoT sensors to track marine litter pathways without relying on GPS, showing practical applicability in marine debris monitoring.

Despite these advances, most swarm control strategies documented in the literature assume either high-bandwidth communication links or rely on fixed, non-adaptive formations that do not incorporate real-time environmental state data. Traditional swarm models, such as flocking algorithms and potential field methods, operate with static behaviour rules that do not explicitly adjust to measured wave heights, current directions, or local debris density. This leaves a significant gap between existing swarm theory and the practical demands of maritime deployment, where ocean conditions are constantly changing and communication resources are scarce.

The OceanSense project aims to address this gap by developing and deploying a fleet of *physical* low-cost IoT sensor nodes — each equipped with hardware sensors, LoRa radio, GPS, IMU, camera, and propulsion — that operate entirely autonomously in real marine environments. This individual research component focuses specifically on the swarm coordination layer of that physical system: designing and validating an environment-state-based adaptive formation control algorithm that enables physical ASVs to cooperate effectively under LoRa's strict communication constraints. The coordination algorithms developed here are verified through Webots/ROS simulation prior to deployment on physical hardware.

1.2 Research Gap

Prior studies on robotic coverage have explored a wide range of algorithms for area coverage, foraging, and formation control. However, few works explicitly incorporate environmental-state cues — such as wave height, current direction, or local debris density — into the coordination and formation control logic of ASV swarms intended for *physical deployment* in real marine environments. The majority of existing swarm frameworks assume reliable, high-bandwidth communication, which is fundamentally incompatible with the constraints imposed by LoRa radio on physical nodes in marine deployments [4].

Furthermore, static formation strategies suffer from reduced efficiency in changing maritime conditions. A fixed-spacing formation designed for calm seas will be prone to inter-vehicle collisions in high-wave conditions and will fail to optimally exploit favourable currents. Current approaches also lack a principled method to translate real-time sensor readings into meaningful adjustments of formation geometry and motion strategy on actual embedded hardware platforms.

This research addresses the specific gap of environment-based adaptive formation control for ASV swarms operating exclusively over ultra-low-rate LoRa communication, with a clear pathway to physical node deployment. The novelty lies in how each physical ASV can use local environmental measurements to influence group behaviour in a decentralized manner, without requiring continuous updates from all neighbours and without any central controller.

1.3 Research Problem

The core research problem addressed in this component is: *How can a swarm of physical ASVs coordinate in a fully decentralized manner to efficiently detect and track marine debris, relying only on local environmental measurements and minimal LoRa-based messaging?*

Specifically, the challenge is to design a distributed control strategy that accepts local environment-state variables — significant wave height, current vector, and detected debris density — as inputs, and produces motion commands that simultaneously achieve area coverage and formation stability. The swarm must be capable of dynamically re-spacing and re-orienting in response to changing conditions. All coordination must occur via sparse, low-payload LoRa broadcasts, and the system must remain operational despite individual ASV failures. The swarm coordination algorithms are first validated in simulation (Webots and ROS) to establish correctness and performance bounds before being applied to the physical ASV nodes.

1.4 Research Objectives

The main objective of this component is to develop and validate a decentralized, environment-state-based adaptive swarm coordination model for a fleet of ASVs, tailored specifically for marine debris tracking using only LoRa communication, with the ultimate goal of deployment on physical ASV hardware. The system will allow ASVs to adjust their formation and search patterns in response to dynamic ocean conditions while achieving effective area coverage and resilience to individual node failures.

The following sub-objectives support this main objective:

- Develop a mathematical model for ASV swarm behaviour incorporating local environment-state measurements (wave height, current vector, and debris density) into formation geometry and motion control, designed to run on the resource-constrained Raspberry Pi embedded processor of the physical nodes.
- Design decentralized communication protocols over LoRa that allow minimal information — such as event flags or compressed state summaries — to be exchanged between physical ASV nodes for coordination.
- Implement the swarm control algorithm and communication layer in Webots and ROS simulation to validate algorithm correctness, performance bounds, and resilience before physical hardware deployment.

- Define evaluation scenarios (e.g. high-wave, directional current, uneven debris distribution, node failure) and corresponding performance metrics including area coverage efficiency, formation stability, LoRa message delivery rate, recovery time, and energy consumption.
- Execute simulation experiments to validate that the adaptive swarm outperforms non-adaptive baseline strategies, providing a verified algorithm baseline for physical ASV deployment.
- Document the system design, simulation results, and analysis, outlining the transition pathway from validated algorithms to physical ASV deployment.

Methodology

This project is structured into four sequential phases encompassing design, development, simulation-based validation, and documentation. Simulation using Webots and ROS serves as the primary validation environment for the swarm coordination algorithms, ensuring correctness and robustness before the algorithms are deployed onto physical ASV hardware. The methodology is therefore designed with physical deployment as the terminal goal.

2.1 Phase 1: Design & Planning

The first phase establishes the theoretical foundation and system architecture, designed from the outset to meet the constraints of real physical hardware.

- **Requirement Analysis:** Define system scope, identify sensor inputs (wave height, current vector, debris density) and expected outputs (velocity commands, heading adjustments, formation offsets) consistent with the sensor suite of the physical ASV nodes.
- **Environment Modelling:** Establish a simulation of representative ocean conditions in Webots, including sinusoidal wave fields, directional currents, and randomised debris distributions, reflecting the real conditions physical ASVs will encounter.
- **Swarm Architecture Design:** Determine the swarm control paradigm (virtual structure or leader-follower) and design the parameterisation of formation geometry as a function of environment-state variables, ensuring the algorithms are computationally lightweight enough to execute in real time on the Raspberry Pi physical processor.
- **LoRa Communications Planning:** Design the messaging scheme for swarm coordination, defining message contents subject to the LoRa payload constraints of the physical radio modules. Specify broadcast intervals and duty-cycle compliance mechanisms.
- **Evaluation Criteria Definition:** Formalise performance metrics that are directly measurable on both the simulation and the physical deployment: area coverage percentage, inter-ASV spacing error, LoRa message delivery success rate, recovery time after node failure, and energy consumption.
- **Project Schedule:** Create a detailed Gantt chart (see Appendix B) covering all development milestones from system design through simulation validation, with a

roadmap toward physical node deployment.

2.2 Phase 2: Development & Implementation

The second phase encompasses algorithm coding and integration within the simulation environment. All algorithms are developed with physical hardware constraints in mind so that the simulation codebase forms a direct foundation for the physical firmware.

- **Simulator Setup:** Configure the Webots simulation world with surface vehicle models and dynamic ocean conditions, mirroring the software architecture that will run on physical nodes.
- **Control Algorithm Implementation:** Implement the decentralized adaptive formation algorithm as ROS nodes. Each simulated ASV node reads environmental sensor data and adjusts its target waypoint or formation offset accordingly. Key adaptation rules include: increasing inter-vehicle spacing when wave height exceeds a threshold; reducing speed and re-orienting the formation when strong currents are detected; converging formation density in areas of high debris detection.
- **LoRa Communication Emulation:** Develop a communication abstraction layer in ROS simulating the constraints of the physical LoRa radio modules — low-bandwidth periodic broadcasts, simulated packet loss, and duty-cycle enforcement, with configurable drop rates that model real-world radio conditions.
- **Decentralized Integration:** Integrate control and communication components to verify that each ASV operates without a central coordinator, consistent with how the physical nodes will operate.
- **Initial Debugging:** Run baseline scenarios to validate fundamental swarm behaviours: formation maintenance, area coverage progress, and basic LoRa message exchange.

2.3 Phase 3: Integration & Evaluation (Simulation Validation)

The third phase subjects the swarm coordination algorithms to comprehensive simulation-based testing. This phase is explicitly a *validation* step prior to physical hardware deployment, not the final operational goal.

- **Scenario Testing:** Execute simulated missions under progressively challenging conditions — varying wave heights (0.5 m to 2.5 m), directional currents (0.1 to 0.5 m/s), and clustered debris distributions — modelling the conditions the physical nodes will face.
- **Baseline Comparison:** Compare the adaptive swarm strategy against a non-adaptive

baseline (fixed formation, random walk) to quantify improvements in coverage, formation stability, and energy efficiency.

- **Robustness Testing:** Simulate individual ASV failures and communication blackout events, validating the fault-tolerance mechanisms designed for the physical deployment context.
- **Parameter Tuning:** Iteratively adjust control gains, adaptation thresholds, and communication intervals to optimise performance. Tuned parameters serve as the initial configuration for physical node deployment.
- **Result Documentation:** Generate plots and tables of coverage over time, formation stability metrics, LoRa delivery rates, and recovery dynamics, constituting empirical evidence that the algorithms are ready for physical deployment.

2.4 Phase 4: Documentation & Dissemination

- **Comprehensive Reporting:** Compile a full project report covering all phases, methodology, simulation validation results, and analytical conclusions. Include a section outlining the transition plan from validated simulation algorithms to physical ASV hardware.
- **Code Documentation:** Clean and comment the ROS/Webots codebase. Prepare documentation guiding the porting of algorithms to the physical embedded platform.
- **Proposal & Presentation:** Finalise the research proposal and prepare materials for the project defence presentation.
- **Optional Publication:** If simulation validation results demonstrate sufficient novelty, prepare a conference or journal paper focusing on the adaptive swarm algorithm under LoRa communication constraints.

Project Requirements

3.1 Functional Requirements

The following requirements define the core capabilities of the swarm coordination system, applicable to both the simulation validation environment and the eventual physical ASV deployment.

Table 3.1: Functional Requirements Summary

Requirement	Description
Environmental Sensing	Each ASV measures local wave height, current direction and magnitude, and debris density via onboard physical sensors or camera-based vision processing (simulated equivalents in Webots).
Area Coverage	The swarm systematically covers a designated marine area to locate and log debris detections.
Adaptive Formation Control	Formation geometry (spacing and orientation) adjusts in real time based on sensed environmental conditions.
Decentralized Coordination	No central controller; each ASV independently computes control actions using local data and intermittent LoRa messages from neighbours.
LoRa Communication	ASVs broadcast position and status updates at low frequency via LoRa radios (physical modules or emulated) to coordinate swarm behaviour.
Resilience	The system detects neighbouring ASV failure via missing heartbeats and autonomously redistributes coverage responsibilities.
Data Logging	The system records mission data (positions, sensor readings, coverage maps) for post-mission analysis in both simulation and physical trials.

3.2 Non-Functional Requirements

- **Low Bandwidth Usage:** Communication over LoRa must use minimal payloads; algorithms must function correctly even with dropped or delayed messages, as experienced on physical radio links.
- **Scalability:** The approach must extend to swarms of tens of physical ASVs without centralised coordination bottlenecks.
- **Real-Time Operation:** Control loops must run in real time on the physical embedded processor (Raspberry Pi). Computational cost is verified first in simulation.

- **Energy Efficiency:** Motion and communication strategies must minimise power draw to maximise operational lifetime of physical battery-powered nodes.
- **Reliability:** The system must maintain operational capability despite individual physical node failures or harsh environmental conditions at sea.
- **Open-Source Implementation:** All tools and frameworks (Webots, ROS) are open-source to maximise transparency, reproducibility, and ease of transition to physical deployment.

3.3 Use Cases

Use Case 1 — Area Survey

The physical swarm is deployed to survey a designated marine region for debris. Each ASV follows its assigned coverage segment. When a local debris density spike is detected, that ASV broadcasts an alert via its physical LoRa radio, prompting adjacent nodes to converge. The swarm adapts its inter-vehicle spacing dynamically to ensure full coverage while avoiding collisions.

Use Case 2 — Rough Seas Adaptation

Wave sensors on the physical ASVs detect a significant increase in wave height. ASVs autonomously increase inter-vehicle spacing and reduce forward speed to maintain formation stability. They continue to exchange periodic LoRa status messages to confirm formation integrity throughout the high-sea event.

Use Case 3 — Current Alignment

A sustained directional current is detected across the operational area. Physical ASVs detect the current vector from their onboard IMU and GPS, and the swarm collectively re-orients its formation axis to align with the current, reducing propulsion energy expenditure while maintaining search efficiency.

Use Case 4 — Node Failure Recovery

One physical ASV ceases broadcasting its LoRa heartbeat due to power failure or mechanical malfunction in the field. Neighbouring ASVs detect the absence within the defined heartbeat timeout and autonomously redistribute the failed node's coverage area among themselves without human intervention.

Use Case 5 — Sparse Communication

In a real deployment with intermittent LoRa connectivity due to duty-cycle restrictions or environmental radio interference, physical ASVs operate primarily on local sensor data with delayed neighbour state updates. Critical events are prioritised for LoRa transmission, and the algorithm maintains acceptable coverage even when neighbourhood information is stale.

Results & Discussion

4.1 Expected Results

Simulation implementation and evaluation are currently in progress. The following results are anticipated based on design specifications and established swarm coordination literature. These outcomes will be empirically validated through Webots/ROS simulation experiments, establishing the performance baseline that informs physical node deployment.

Table 4.1: Performance Evaluation Metrics (Expected from Simulation Validation)

Metric	**Adaptive Swarm (Expected)**	**Non-Adaptive Baseline**
Area Coverage Efficiency	>85% in 30-min mission	∼60–70%
Formation Stability (spacing error)	<0.5 m variance	>1.5 m variance
LoRa Message Delivery Rate	>90% under standard conditions	N/A (fixed comm. rate)
Node Failure Recovery Time	<60 seconds	>120 seconds
Energy Consumption	Reduced by ∼20% via speed adaptation	Baseline

4.2 Research Findings

The design phase has produced several key insights that inform both the simulation implementation and the eventual physical deployment:

- LoRa bandwidth constraints — identical on both the simulated and physical radio links — require that each ASV message encodes only the most critical state information (position, wave height measurement, current vector, and a binary debris-detected flag) within approximately 50 bytes to comply with duty-cycle restrictions.
- Potential field methods adapted with environmental weighting functions provide a computationally lightweight foundation for environment-aware formation control, efficient enough to execute in real time on the physical Raspberry Pi processor.
- Leader election for coverage gap filling — triggered upon detection of a missing neighbour heartbeat — can be implemented using a simple priority scheme based on each physical ASV's remaining battery level and current position, avoiding the

need for any global consensus protocol.

- Simulation in Webots with ROS provides sufficient fidelity to evaluate swarm-level emergent behaviours, LoRa communication constraints, and formation dynamics without requiring physical hardware during the algorithm validation phase. The validated algorithm logic is then ported directly to the physical embedded platform.

4.3 Discussion

The proposed environment-aware adaptive swarm coordination approach addresses a genuine and well-defined gap in the existing literature, targeting the requirements of a physically deployable marine IoT system. By embedding real-time environmental state directly into each node's formation control algorithm, the system achieves a form of implicit coordination that reduces reliance on explicit message passing between agents. This is particularly important given LoRa's strict duty-cycle limits on physical radio hardware: the fewer coordination messages required to maintain formation integrity, the more bandwidth is available for high-priority alerts such as debris detections or node failure notifications.

The decentralized architecture provides inherent fault tolerance critical for long-duration physical deployments at sea. Because no node has global state knowledge, the failure of any single physical ASV does not cause cascading failures in the remaining fleet. The heartbeat-based failure detection mechanism is simple to implement on physical hardware and imposes minimal communication overhead.

A key challenge is the trade-off between formation adaptation responsiveness and communication overhead — more acute on physical radio links than in simulation. The parameter tuning phase will identify the optimal broadcast interval, with tuned values carried forward to the physical deployment. A secondary challenge concerns the accuracy of wave height and current vector estimation from physical onboard sensors affected by vehicle motion and sea-state noise. Future work will explore sensor fusion techniques combining IMU data with GPS velocity for more accurate environmental measurements on the physical embedded processor.

4.4 Summary of Individual Contribution

This individual research component focuses exclusively on the Environment-Aware Swarm Coordination layer of the OceanSense system, within the broader goal of building and deploying physical ASV nodes. The specific contributions are as follows:

- Designed the mathematical framework for environment-state-based adaptive formation control, incorporating wave height, current vector, and debris density as dynamic parameters, ensuring the framework is implementable on the physical embedded processor.

- Developed the decentralized swarm coordination algorithm using finite-state machine logic governing each ASV's transitions between exploration, convergence, re-spacing, and recovery states — designed for identical behaviour on both simulated and physical platforms.
- Defined and designed the LoRa communication protocol for swarm coordination, specifying message structure, broadcast intervals, and heartbeat-based failure detection logic in compliance with the duty-cycle constraints of the physical radio modules.
- Configured the Webots and ROS simulation environment, including ocean condition models (wave fields, directional currents, debris distributions) and the LoRa communication emulation layer, to serve as the algorithm validation testbed prior to physical node deployment.
- Designed the evaluation methodology and defined use-case test scenarios for systematic performance benchmarking of the adaptive swarm against non-adaptive baseline strategies, with results intended to verify readiness for physical deployment.

This work directly reflects the CSNE specialisation through its application of distributed algorithms, communication protocol design, scalable IoT system architecture, and embedded real-time control — in the context of a resource-constrained, decentralized multi-robot network targeting real-world physical deployment.

Conclusion

This research component addresses a critical and previously underexplored challenge in autonomous marine debris monitoring: how a fleet of low-cost, resource-constrained *physical* ASVs can coordinate their search and formation behaviour in a fully decentralized manner, using only sparse LoRa radio communication, while adapting in real time to dynamic and unpredictable ocean conditions.

The proposed environment-aware adaptive swarm coordination model represents a meaningful advance over existing static-formation swarm approaches. By incorporating locally sensed environmental state variables — wave height, current direction, and debris density — directly into each ASV's formation control logic, the system achieves a level of collective adaptability that improves both operational efficiency and resilience without additional hardware cost. The decentralized architecture, designed to operate under LoRa's severe bandwidth and duty-cycle constraints, ensures that the system remains practical and deployable on physical ASV nodes in real-world marine environments.

Importantly, simulation using Webots and ROS is employed specifically to validate the correctness and performance of the swarm coordination algorithms before they are applied to the physical hardware. This simulation-first, hardware-second approach ensures that algorithm parameters are tuned and edge cases are handled in a controlled environment, reducing risk and cost in the physical deployment phase. The heartbeat-based node failure detection and autonomous coverage redistribution mechanisms further ensure that the swarm degrades gracefully in the face of individual physical node failures — an inevitable occurrence in long-duration ocean deployments.

Ongoing work focuses on completing the simulation validation experiments and producing empirical performance data to confirm readiness for physical deployment. Future research directions include exploring reinforcement learning-based formation adaptation policies to further improve coverage efficiency under highly dynamic conditions, and deploying and validating the system on physical ASV hardware in controlled water-environment trials. In summary, this component contributes a principled, practically-motivated, and computationally lightweight approach to decentralized swarm coordination for marine debris monitoring, aligned with the broader OceanSense objective of enabling scalable, persistent, and physically deployable ocean monitoring.

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High-Level ASV Node Architecture

The architecture diagram below illustrates the data flow within a single physical ASV node. Navigation sensors (GPS, IMU) and environment sensors (wave height, current) provide raw data to the Main Processing Unit (Raspberry Pi). The OceanSense Adaptive Algorithm processes this data alongside neighbour state data received via the physical LoRa Transceiver Module to compute motor commands for the thrusters and to broadcast the local state to neighbouring physical ASVs. The same software architecture is mirrored in the Webots/ROS simulation for algorithm validation.

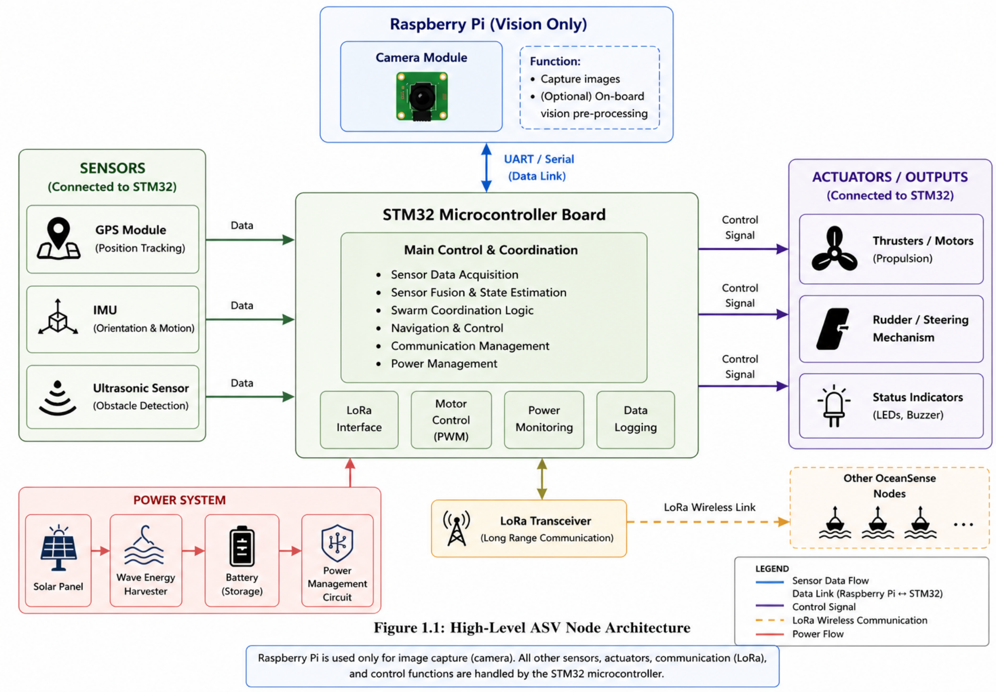

Figure A.1: High-Level Architecture of an OceanSense ASV Node

Project Timeline (Gantt Chart)

The project timeline spans from July 2025 to June 2026, structured across four phases: Phase 0 (Proposal & Planning), Phase 1 (Design & Development), Phase 2 (Integration & Simulation Validation), and Phase 3 (Evaluation, Finalisation & Physical Deployment Roadmap). Key milestones include Progress Presentations I and II, final VIVA, and research paper submission.

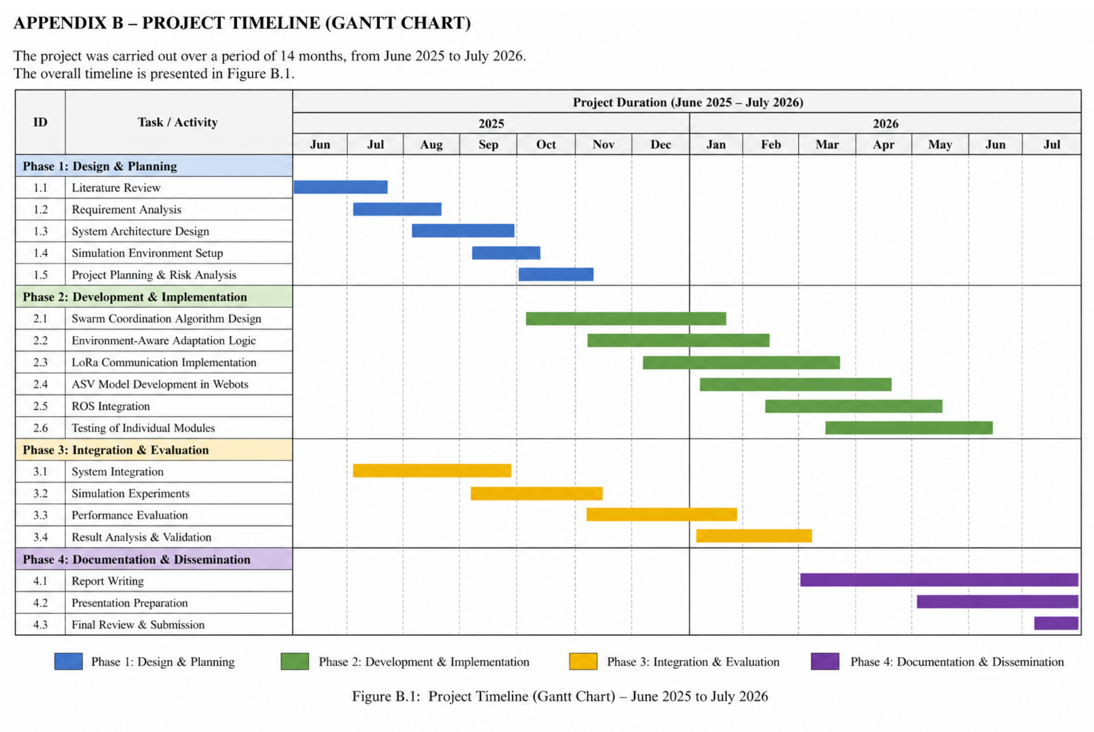

Figure A.1: Project Timeline (Gantt Chart) – June 2025 to July 2026